



When machine learning meets econometrics: Can it build a better measure to predict multidimensional poverty and examine unmeasurable economic conditions?

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ABSTRACT

Poverty is as old as human civilization, hard to eradicate, multidimensional, and difficult to measure. The methods used to measure poverty today are costly, labor-intensive, and time-consuming. Therefore, policymakers find it difficult to target policies when putting poverty reduction initiatives into action. Indigenous communities are among the most disadvantaged and vulnerable populations in society. Their socioeconomic situations are complex and multifaceted. While research on poverty is usually generic, prone to large sampling errors, and intended to guide national policy, research on indigenous people is qualitative. Thus, to measure multidimensional poverty with disaggregated techniques, this work blends machine learning and econometrics. Researchers who have been studying poverty worldwide can replicate all of the approaches, strategies, and resources used in this study. With the best R-square and accuracy, random forest models perform better than all regressors and classifiers combined. It also confirms the causal relationship and current econometric association between multidimensional characteristics and poverty consequences. This study demonstrates the viability of using machine learning to predict poverty in a way that can save costs, cut labor, and maximize time to empower indigenous communities and alleviate the poverty of impoverished societies in the poorest region of Luzon, Philippines.

Video and Presentations to this article can be found online at <https://doi.org/10.1016/j.sctalk.2024.100387>.

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Figures and tables

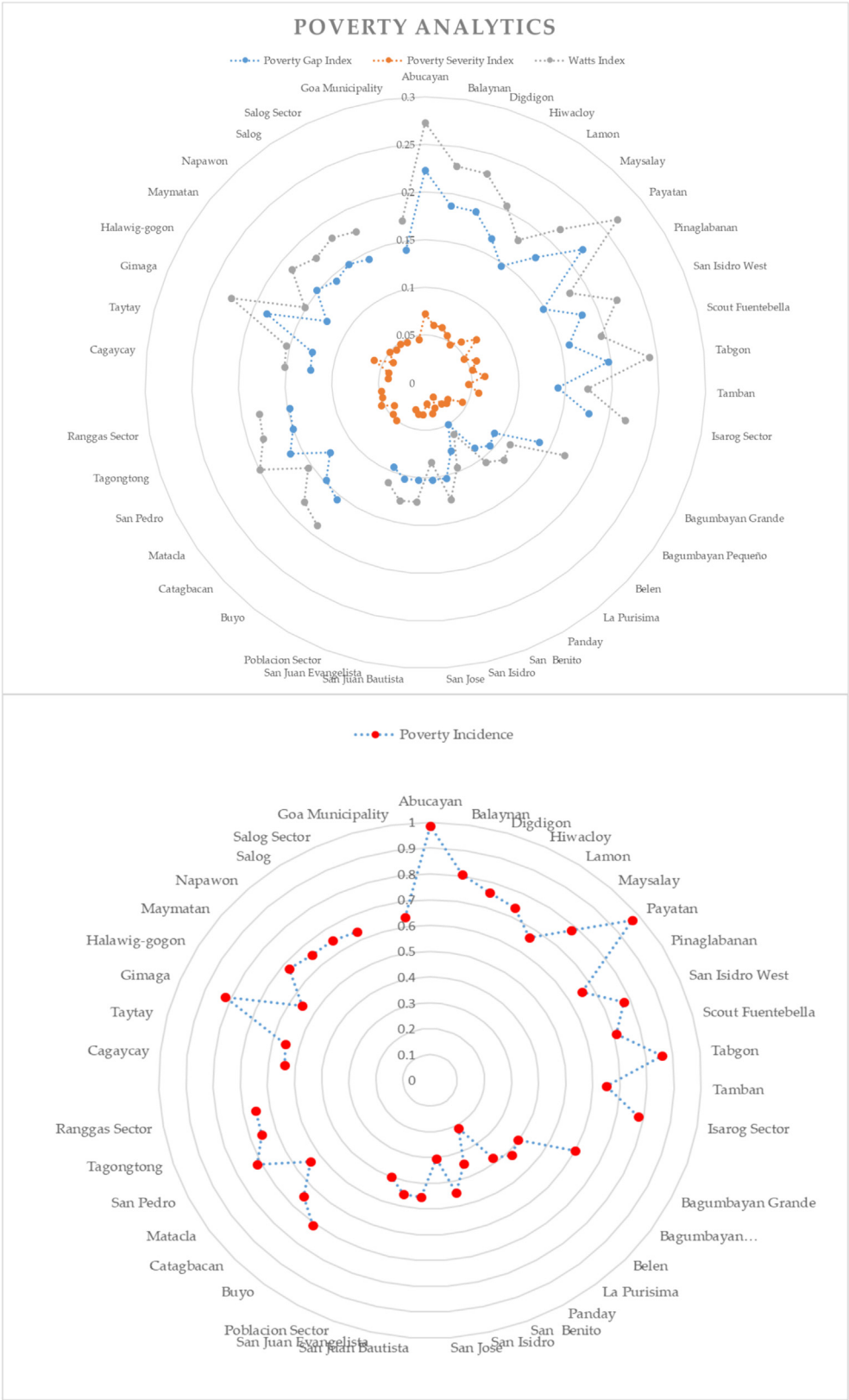


Fig. 1. Comprehensive poverty analytics at disaggregated approaches.

Abucayan is the most impoverished locality. San Jose, San Benito, and Panday have the fewest impoverished households. While poverty is widespread in the Isarog Sector, it is not as severe in the Poblacion community. Watt's Index claims that Isarog communities have a greater prevalence of poverty. Poor households and people are reflected through poverty incidence, which represents the headcount ratio of those living below the poverty line and food thresholds in the Philippines, relative to the total number of households and population [1,2,11,18,21].

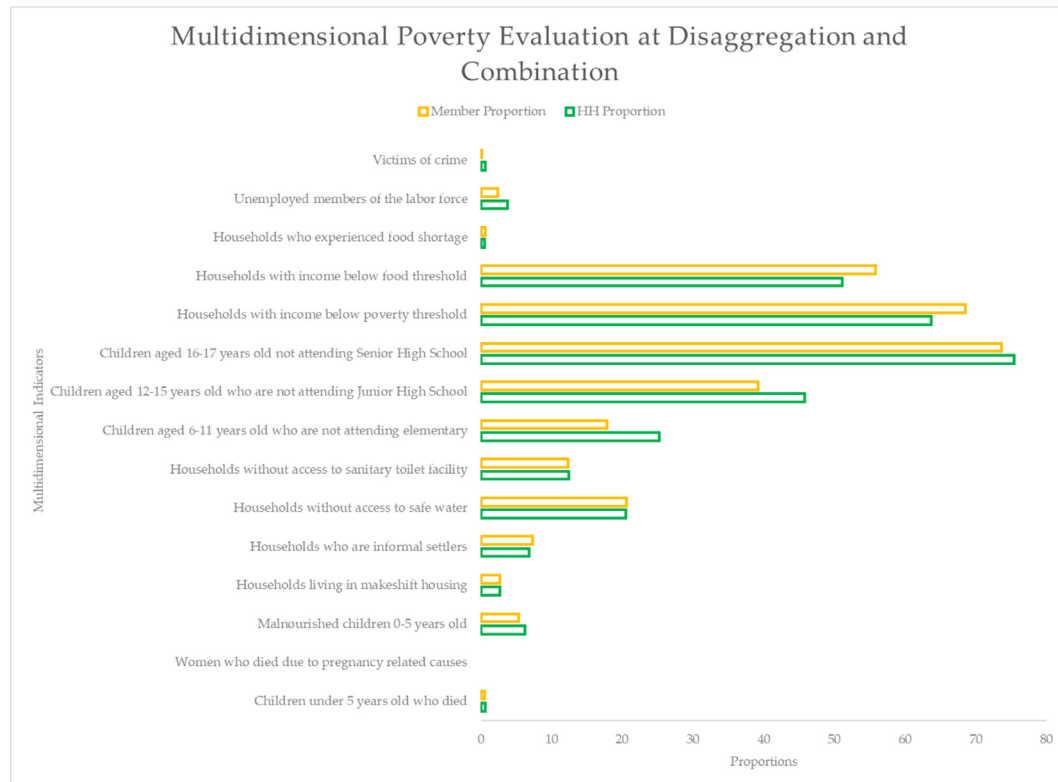


Fig. 2. Results of multidimensional poverty evaluation at disaggregation and combination through the multiple constructs and indicators.

About 63.70% of people live in poverty and 51.10% of individuals do not have access to food. The dynamics of education, income, and health all clearly demonstrate poverty. In the Philippines, households with a monthly income of PhP10,481 (about 186.18 USD) and below for a family of five are considered poor. It is important to consider that different regions may have different poverty lines, so it is pre-requisite to identify the poverty line in your area before applying our suggested ML models. Multidimensional Poverty deprivation refers to the 15 economic variables that we have employed, which serve as predictors or independent variables in our econometric and machine learning models. It offers a more comprehensive approach. Instead of using an index, we employ binary and dummy data to capture non-linear relationships [1,2,21,23].

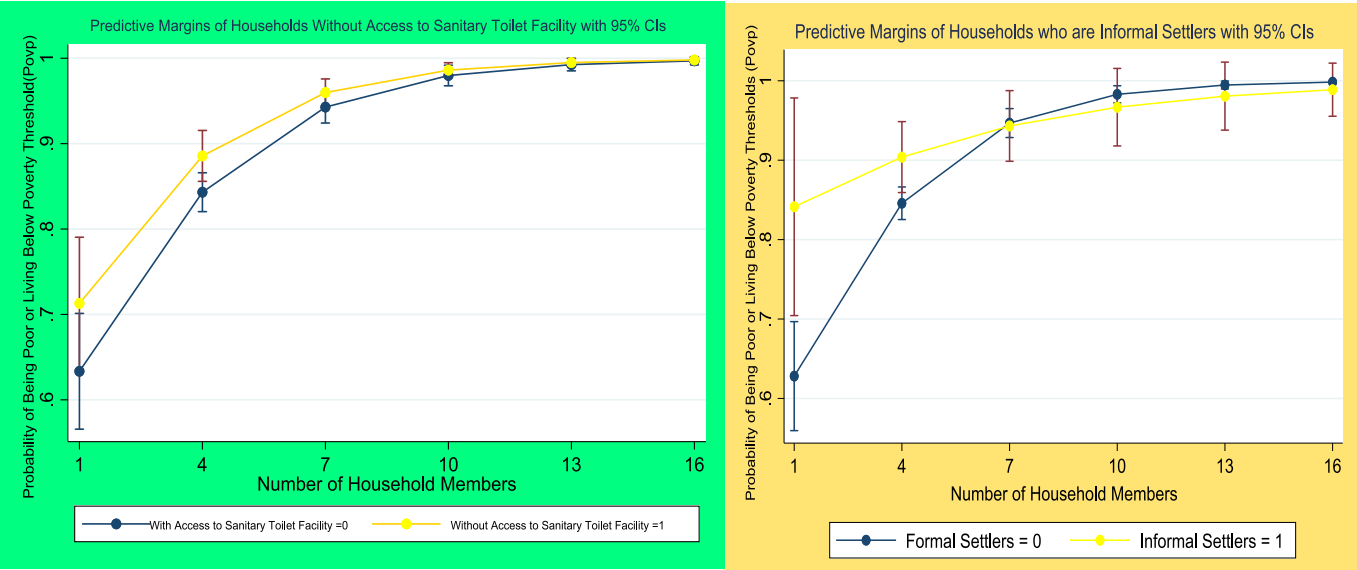


Fig. 3. Classification results and predictive margins of multidimensional variables.

As a factor impacting health dynamics, the results indicate that household size is a strong predictor of poverty. Poverty is more likely to be experienced by informal settlers than by formal settlers. Furthermore, the chance of falling into poverty increases with the number of household members [1,2,6,20,24].

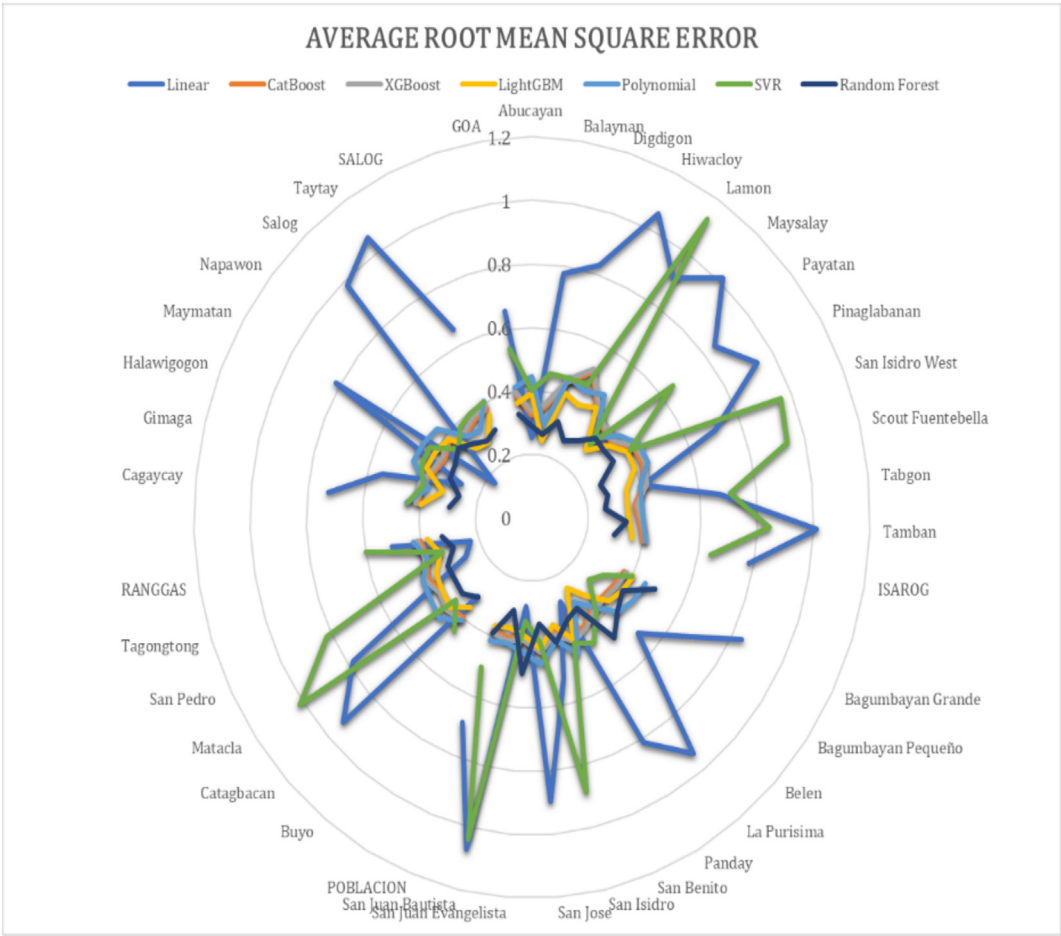


Fig. 4. Average root means square error of machine learning regressors for impoverished societies.

With an RMSE of 0.3298, the random forest method performs better than the others, LightGBM at 0.3642 and XGBoost at 0.4001. These results are derived from ML models (regressors) displayed on video slides 15 and 17. The reason why random forest performed the best is that it can identify explanatory factors, find nonlinear correlations between explanatory and dependent variables, and estimate feature importance, which provides insight into the features that have the greatest influence on the categorization process. Although the outcome of machine learning (ML) is dependent on the context of the data, results from ML applied to similar datasets and models in other locations would be comparable. Maybe the only things that would change are the predictors. The ML model is sustainable in this way since it requires a large amount of data, and the data we utilized is available in the Philippines every two years after the census. Furthermore, even with cross-sectional data, different regions can apply machine learning once the data is accessible for them, because a medium-term social outcome can be predicted by an ML model [1,2,19,10,15].

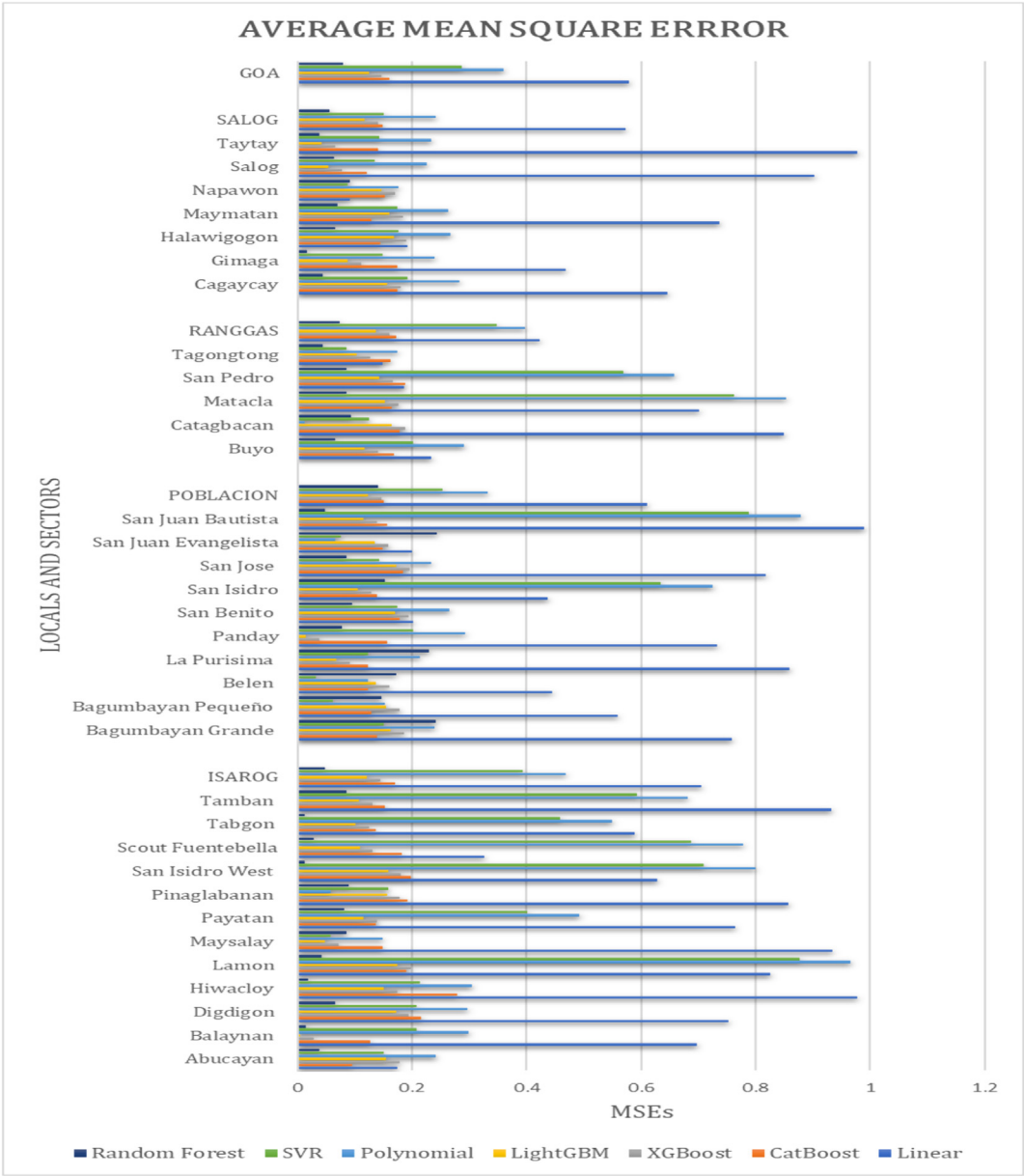


Fig. 5. Average mean square error of machine learning regressors for impoverished societies.

For the entire municipality, the random forest regression algorithm has the lowest mean square error, at 0.0792. XGBoost (0.4722) and LightGBM (0.1242) are in its wake. The ML models we used are specific to the areas we studied. Therefore, we initially employed econometrics to identify proper predictors for ML models that are specific to our local context. However, the poverty models (logistic) and the ML models (regressors and classifiers) are generic and can be applied to other regions or countries. Consequently, repeatability is highly feasible in this regard. Other regions of the world would need to identify their own predictors, but the models we used could also work for them [1,2,15,22].

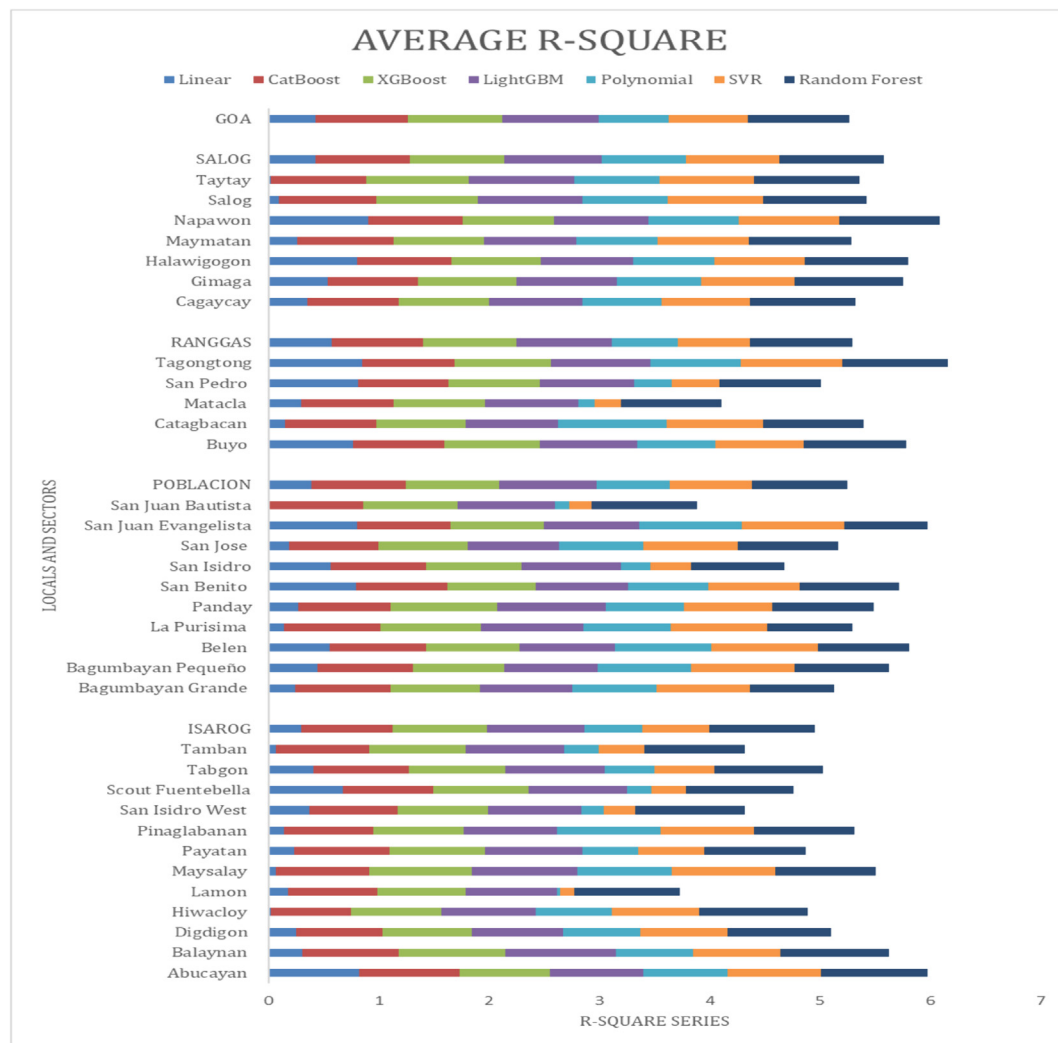


Fig. 6. Average R-square of machine learning regressors for impoverished societies.

With an R-square of 0.4221, the linear regression has the lowest value. The random forest regressor produced the highest R-square, 0.92075. It suggests that the regression that best matches the regression line and has the lowest error score is the random forest regression. Since the data examined in this paper is based on a comprehensive census of all populations including indigenous people, there are no known limits. As a result, it remains exempt to sampling error and upholds the sample assumptions of internal and external validity. The writers thoroughly examine and filter the data's multidimensional variables. The dataset is well-sized and devoid of well-known biases. But time is of the essence. It takes time to create trustworthy and verifiable datasets and models [1,2,12,13,14].

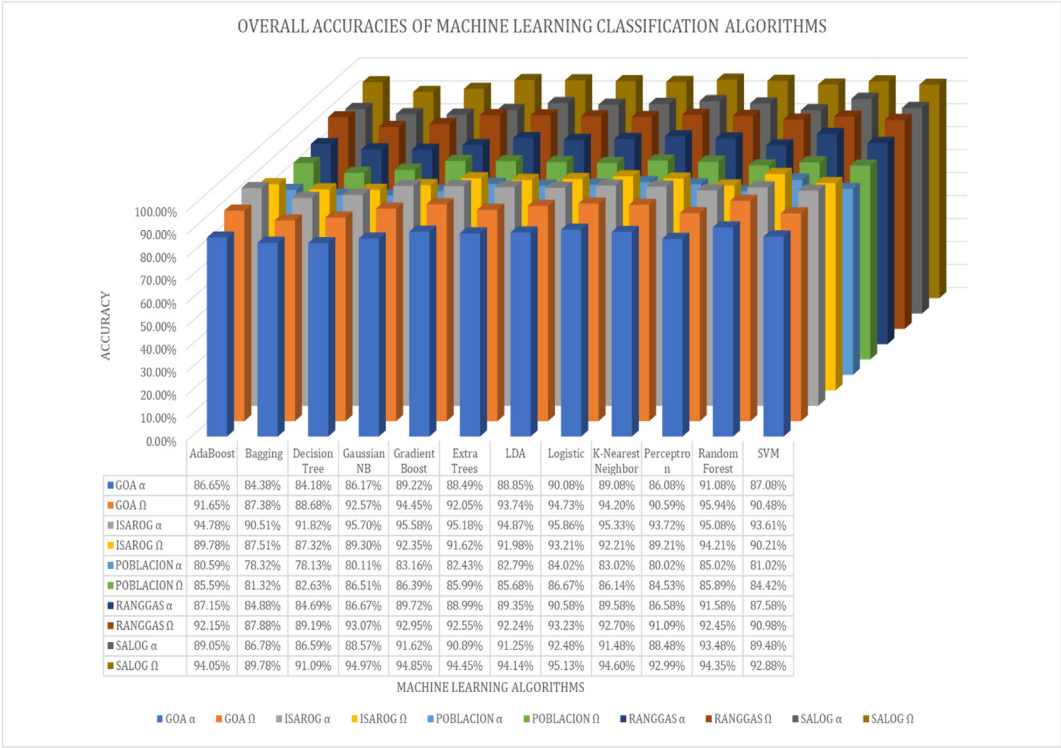


Fig. 7. Performance evaluation of machine learning regressors for impoverished societies.

With an accuracy of 0.9108, the random forest regression classifier has the highest performance. The algorithms with the lowest accuracy are Bagging (0.8437) and Decision Tree (0.8418). The R-squares for the SVM, AdaBoost, Gaussian NB, and Perceptron are 0.86–0.87. The random forest classification has the greatest accuracy rate of 95.94% in pipeline algorithms. Concerning the future improvements, the utilization of econometric models and predictive analytics datasets has the potential to generate empirical data that strengthens poverty theories that apply to entire populations, including indigenous populations. This would contribute to the existing body of knowledge about the dearth of multidimensional poverty metrics in developing nations like the Philippines, especially for indigenous populations. Applying the models to all other tribes in the nation and utilizing artificial neural networks could both enhance forecast accuracy and yield improvements [1,2,8,16,17].

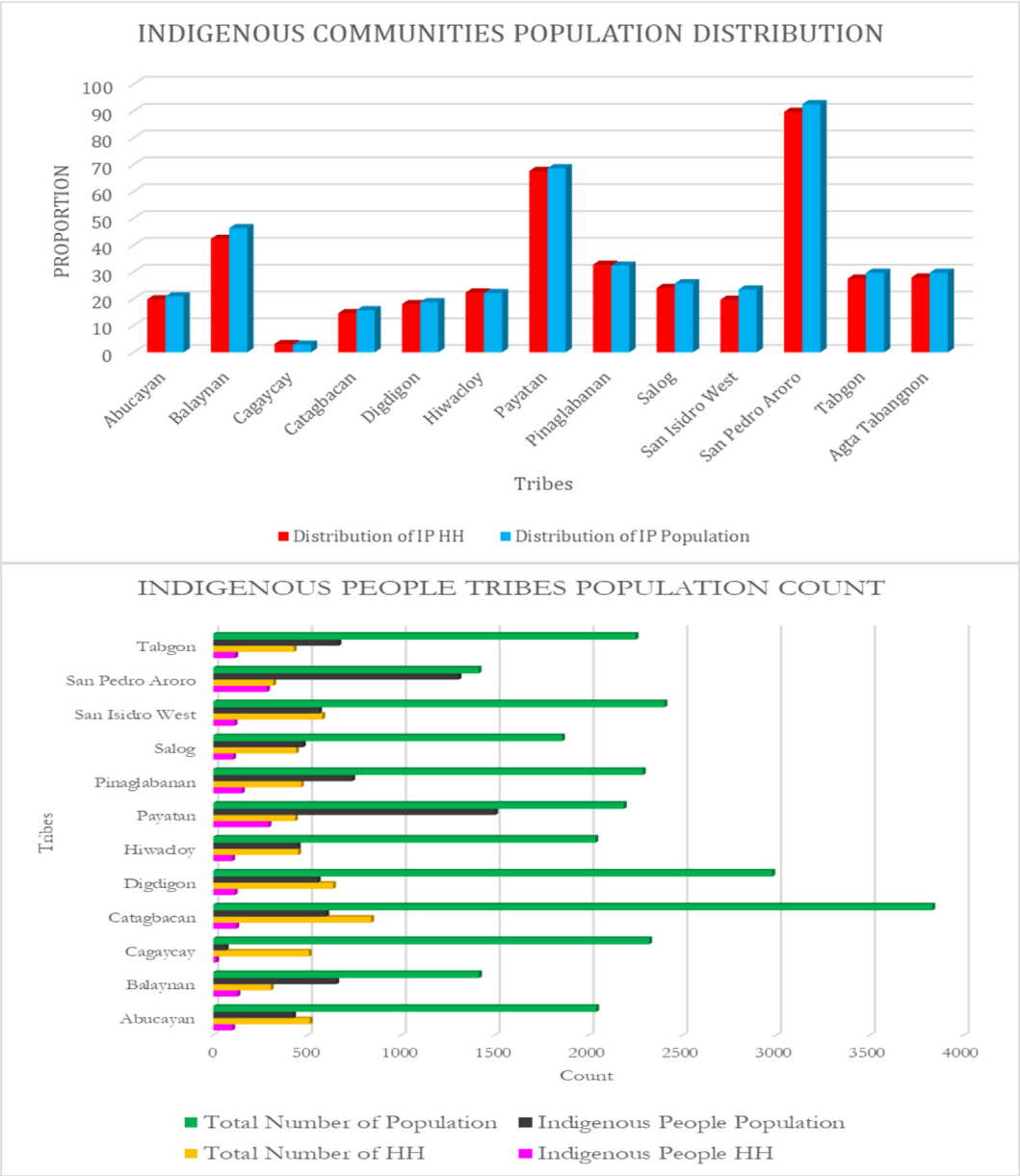


Fig. 8. Indigenous communities' population analytics.

Indigenous people make up about 11.84% of all households in Goa, Camarines Sur. San Pedro Aroro, Payatan, and Balaynan are the most populous indigenous tribes, with respective total population distributions of 92.50%, 68.66%, and 46.30% [3,4,7,11].



Fig. 9. Multidimensional socioeconomic evaluation and poverty examination of indigenous people tribes.

There are 1660 houses of indigenous people and 8022 persons in total. Eighty-two percent of people live in poverty, and 71% do not have access to food. Evaluation results for schooling are largely constant throughout barangays, suggesting that most children in the neighborhood are not enrolled in school or have stopped enrolling as educational levels rise [3,4,7,11,18].

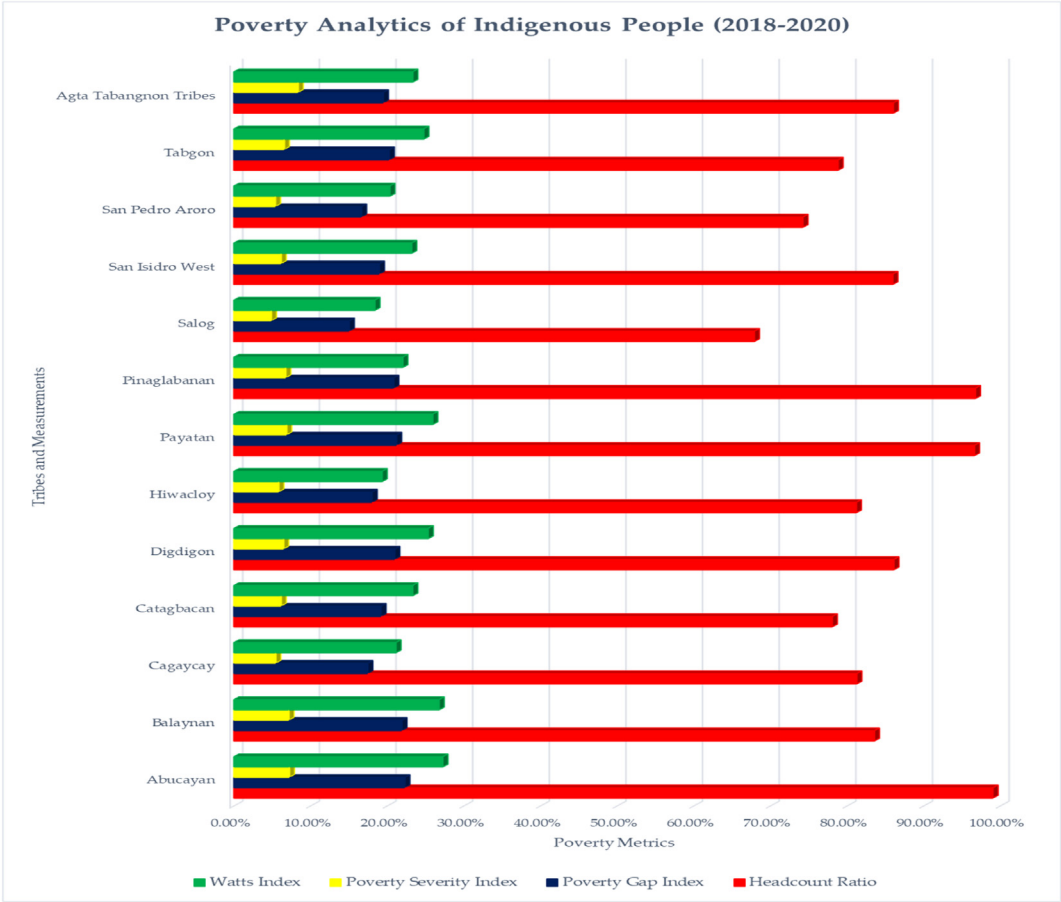


Fig. 10. Indigenous people poverty analytics.

The Poorest tribe is the Abucayan Tribe while the Least Poor tribe is the Salog Tribe [1,3,4,21,23].

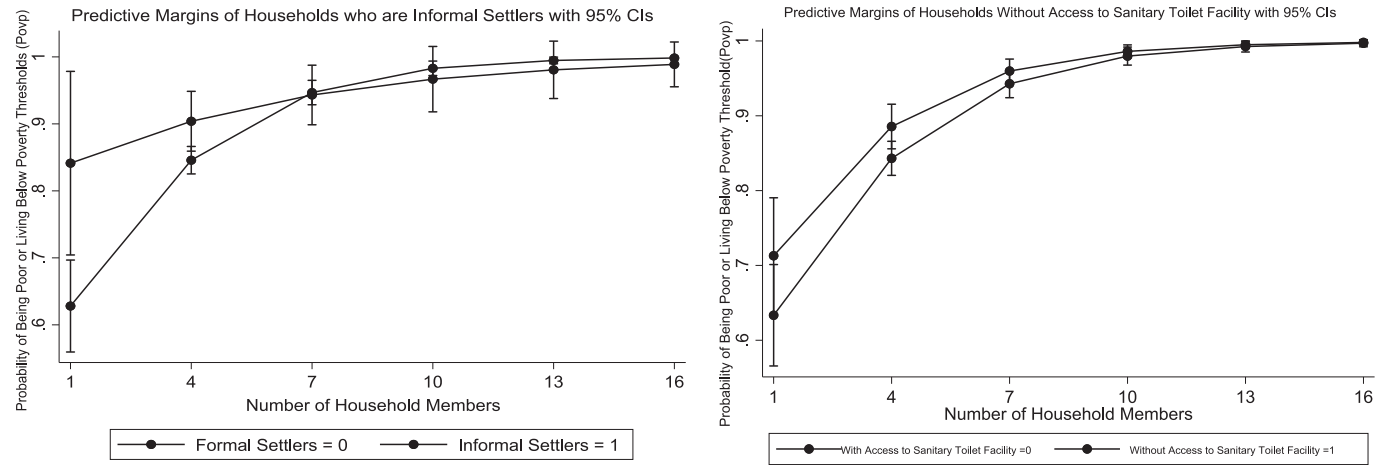


Fig. 11. IP Classification Prediction Margins.

Living below the poverty level is more common among IP informal settlers. Moreover, the probability of poverty increases with the number of household members. The results of the Estat categorization are positive. Overall, 86.14% of cases were appropriately classified [3,4,5,9,25].

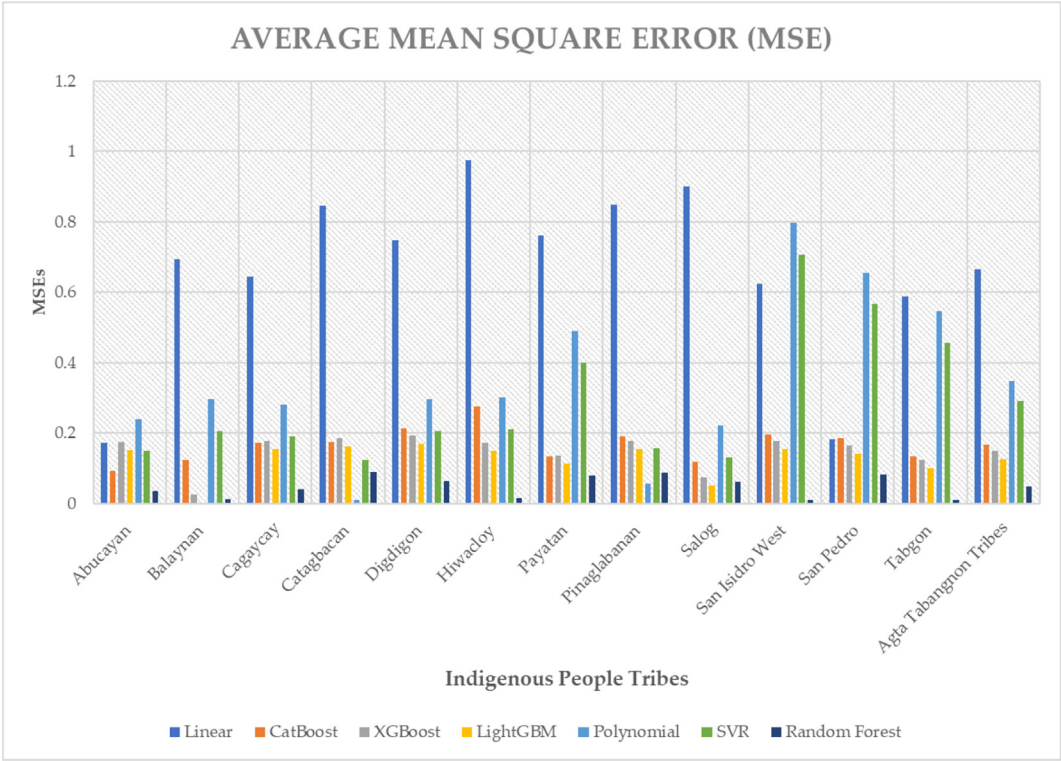


Fig. 12. Average mean square error (MSE) of machine learning regressors in IP modeling.

After LightGBM and XGBoost, the random forest regression method has the lowest mean square error (0.5497). Causality tests using econometrics (logit and probit regression) were initially performed to identify predictors for the ML models. All the multidimensional indicators used in the model (Fig. 9) and reflected in the video have a causal relationship with poverty for IP groups. The following variables have been found to have causality: Health and Nutrition, Housing, Water and Sanitation, Basic Education, Income and Livelihood, and Peace and Order. Since these variables are causally linked to poverty, they serve as predictors in the ML models. Given that these variables have a causal relationship and are predictors of poverty in the models, our policy mapping and targeting analysis focuses on these variables. We concentrate on addressing these variables in order to improve policies, as doing so would alleviate poverty and promote development [3,4,15,22].

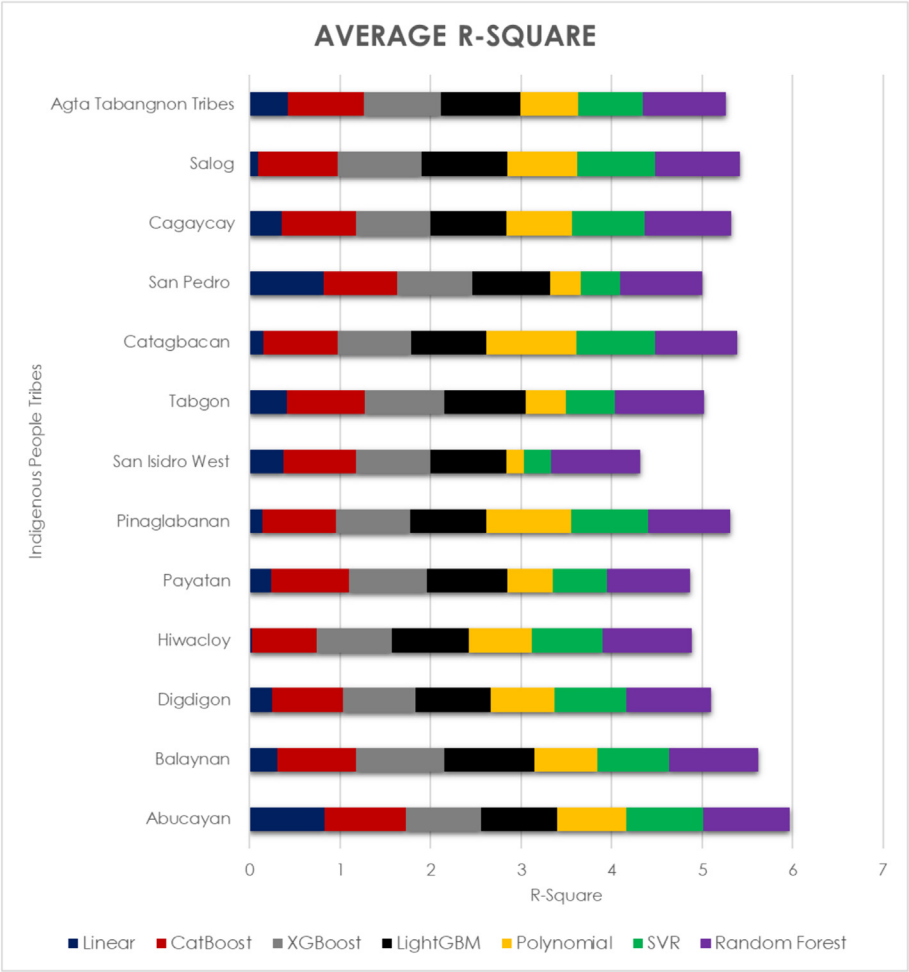


Fig. 13. Average R-square of machine learning regressors for Indigenous Communities.

Of all the techniques, linear regression has the lowest R-square, at 0.4222. Next are a polynomial algorithm and an SVR. With an R-square of 0.9208, the random forest regressor produced the highest results, although CatBoost, XGBoost, and LightGBM all performed similarly. It implies that the random forest regression is the one that most closely resembles the regression line and has the lowest error score [3,4,12,13,14].

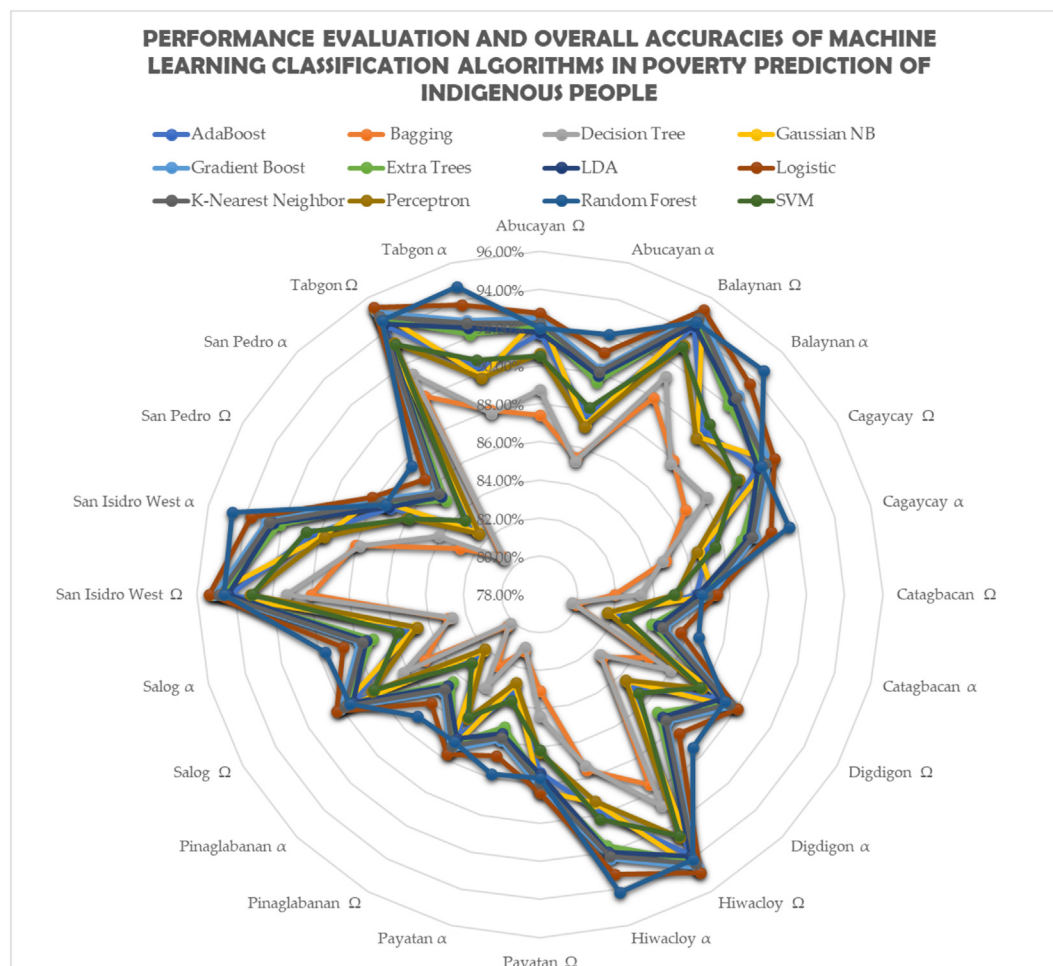


Fig. 14. Performance evaluation of ML classifiers at random and pipeline for IP.

For all tribes at random (α), the random forest regression classifier has the highest accuracy (90.69%); K-Nearest Neighbour, Gradient Boost, and Logistic follow. At pipeline (Ω), the random forest classification has the greatest accuracy rate at 94.89%. The next two classification strategies are gradient boost and logistic. The results affirm that all 15 indicators serve as predictors of poverty, and the random forest algorithm emerges as the most effective model for poverty prediction. Therefore, it is crucial to address all these indicators adequately since they are the root causes of poverty. By formulating policies that target each predictor, poverty can be alleviated, leading to advancements in economic development [3,4,8,16,17].

CRedit authorship contribution statement

Emmanuel A. Onsay: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jomar F. Rabajante:** Writing – review & editing, Validation, Supervision, Software, Resources, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Data availability

We have shared our data publicly. The links to our datasets were provided through the references section (Ref. 1, 2, 3, 4)

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Ethics statement

All graphs and figures presented in this work were produced and visualized by the authors. The utilization of all photos in the video presentation has obtained formal consent from the subject, university, government, and stakeholders. The conduct of this study was approved by Partido State University and the University of the Philippines Los Baños. The Goa, Camarines Sur Local Government Unit gave permission for the results of the processed data to be used, manipulated, analyzed, and communicated. Researchers voluntarily conduct the data analysis and procedures; they are not a component of any experiment. Our human data are secondary, indirect, and socioeconomic in nature; they don't entail any laboratory work. Thus, ethical clearances are not required. Furthermore, this study does not involve any animal testing, direct human volunteers, or data acquired from social media platforms.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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